

UTSAV DARLAMI

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[Website](#) — [Github](#) — [LinkedIn](#)

Graduating Dec 2026 — Seeking Summer '26 Internships & Full-Time

EXPERIENCE

Graduate Research Assistant

Department of Computer Science, University of Alabama in Huntsville

August 2025 - Present

Huntsville, Alabama

- Conducting research on interaction patterns and view management in multi-display environments.
- Developing an intelligent system using **React Flow** and **LLMs** to assist with spatial information organization.

Machine Learning Engineer (Full Stack)

Docsumo

September 2021 - April 2025

Kathmandu, Nepal

- **Lead Developer** for a Python package streamlining LLM-based information extraction using **LiteLLM**, **Langfuse**, and **Pydantic**; this library optimized experimentation and powered the company's fully automated information extraction API.
- Designed and built an LLM-based table extraction pipeline to convert unstructured text into structured data; benchmarked performance against existing solutions using models like GPT-4o, Mixtral-8x7B, and Llama 3.1, achieving **75%** accuracy with **GPT-4o** in production and scaling to process an average of **300+ documents** for 2 enterprise clients.
- **Slashed processing time by 60%** by engineering a data featurization pipeline for specialized **scikit-learn** models (identifying column labels, valid rows) from OCR data.
- Collaborated directly with clients to define requirements and deploy custom extraction logic specific to their business use cases.
- Built and deployed a scalable text classification API using **Fasttext**, **SpaCy**, **FastAPI**, and **GCP**, which currently supports four downstream services.

PROJECTS

Agent Starter Template (Google ADK)

Python, FastAPI, Google ADK, JavaScript

[Github](#)

- Built a stateful AI agent architecture using FastAPI and the Google ADK, including a JavaScript client for real-time state updates.
- Created a tool-calling system with Pydantic to let the AI safely execute functions.
- Implemented a diff engine to send only data changes, reducing bandwidth usage.

Evolutionpy

Python, NumPy, Meta-heuristic Algorithms, OOP

[Github](#)

- Built a Python library for evolutionary optimization to solve complex problems.
- Designed a modular codebase using Object-Oriented Programming (OOP) to easily support custom fitness functions.

Sandstone Microscopic Image Segmentation

Python, Scikit-learn, Random Forest, Image Processing

[Github](#)

- Generated feature sets from microscopic images using filters like Gaussian, Sobel, and Gabor.
- Trained a Random Forest classifier on these features to identify mineral segments (Quartz, Pore, Clay).
- Achieved a **95%** F1-Score (Dice Coefficient) on the segmentation task.

BreakfastScoop

Python, BeautifulSoup, Flask, NLP, Machine Learning

[Github](#)

- Scraped news articles from multiple Nepali websites using a custom BeautifulSoup script.
- Categorized articles into 10 topics using a Naive Bayes classifier with an **81%** F1-score.
- Developed a Flask application that fetches live news using the scraper and categorizes it using the trained model.

SKILLS

Programming Languages: Python, C/C++, Bash, JavaScript

Databases & Tools: SQL, MongoDB, Docker, Git, Postman, Linux, GCP (Cloud Run, Buckets)

Machine Learning & Data Science: PyTorch, Scikit-learn, OpenCV, spaCy, fasttext, Transformers

LLM Frameworks: LangChain, LiteLLM, Google ADK

Languages: English, Nepali, Hindi

TALKS AND PRESENTATIONS

Presentation: “[ANN-GA for Optimization of Multivariate Function](#)” at [NCMA 2022](#), Nepal

June 2022

EDUCATION

Master of Science in Computer Science

University of Alabama in Huntsville, Huntsville, Alabama

August 2025 - December 2026

Bachelor of Science in Computer Science

Kathmandu University, Dhulikhel, Nepal

August 2017 - March 2022